

①

Homework #5

CHE 210, Fall '08
Yergenny Strawnow
(TA)Solutions

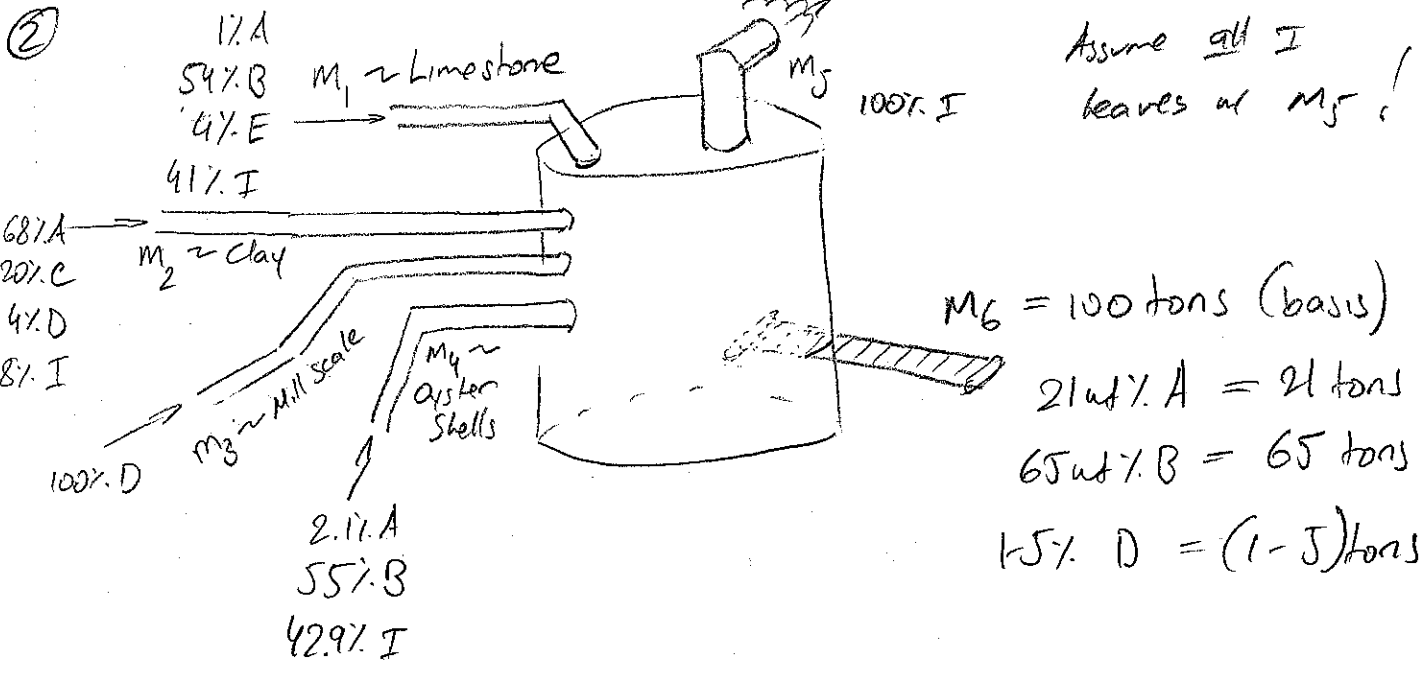
(1) Rework Murphy 2.67. As in class, label species as A, B, C, D, E, & I (SiO_2 , CaO , Al_2O_3 , Fe_2O_3 , MgO , & Ignitable Mat.)

note: Oyster Shell Data has been changed:
- see table below...

From Murphy P. 59 & the changed data in problem statement;

Component in:	Limestone	Clay	Mill Scale	Oyster Shells (wt%)
A	1	68	0	2.1
B	54	0	0	55
C	0	20	0	0
D	0	4	100	0
E	4	0	0	42.9
I	41	8	0	60
Cost (\$/ton)	88	35	free	

②



Species Balances

A: $.01 m_1 + .68 m_2 + .021 m_4 = 21 \text{ (tons)}$

B: $.54 m_1 + .55 m_4 = 65$

C: $.20 m_2 = ? = (\text{wt\% C})_{m_6} \cdot 100$

D: $.04 m_2 + m_3 = 1 \leq (\text{wt\% D})_{m_6} \leq 5$

E: $.04 m_1 = (\text{wt\% E})_{m_6} \cdot 100$

I: $.41 m_1 + .08 m_2 + .429 m_4 = m_5$

overall mass: $m_1 + m_2 + m_3 + m_4 = m_5 + 100$

③

	m_1	m_2	m_3	m_4	m_5	x_c	x_E	b
A	.01	.68	0	.021	0	0	0	21
B	.54	0	0	.55	0	0	0	65
C	0	.20	0	0	0	-100	0	0
D	0	.04	1	0	0	0	0	1475
E	.04	0	0	0	0	0	-100	0
F	.41	.08	0	.429	-1	0	0	0
G	1	1	1	1	-1	0	0	100
m_6	0	0	0	0	0	1	1	1349

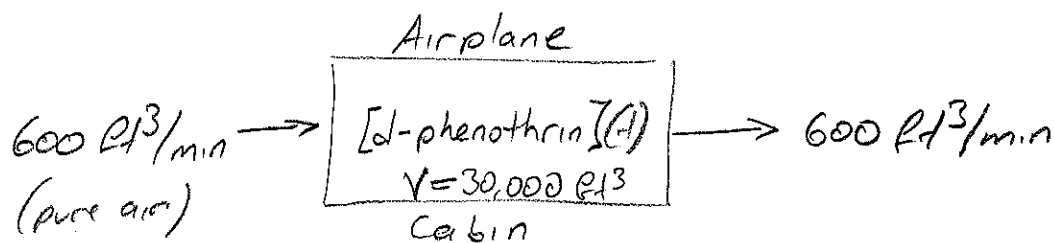
My TI-89 could not solve this matrix! !!

You could try using Maple or Mathematica, but the new data may have rendered the system unsolvable...

b. It does not appear possible to resolve these equations by successive substitutions, as if you look closely at the matrix above, you can see that it seems impossible to isolate any single variable to begin the process...

①

(2) Murphy 3.20



@ $t = t_0$ (start of process), we have $\frac{10^4 \text{ parts d-phen}}{10^{12} \text{ parts air}} \times 30,000 \text{ ft}^3 \text{ air}$
(=0)

$$\therefore [d\text{-phen}]_{t_0} = 300 \text{ ft}^3 \text{ d-phen}$$

Assuming that "part" is a volumetric measure, we

can leave everything in given units & not convert to moles. This is because volume scales w/ moles.

@ $t = t_f$ (end of process), we have $\frac{100 \text{ parts d-phen}}{10^{14} \text{ parts air}} \times 30,000 \text{ ft}^3 \text{ air}$

$$\therefore [d\text{-phen}]_{t_f} = 3 \text{ ft}^3 \text{ d-phen}$$

We can write a Diff. E. Q. to model the process:

But first, we assume that the cabin air is well-circulated. This means that the gas mix that comes out with the exiting stream has the same conc. of d-phen as that throughout the rest of the cabin at that instant in time... (the old "well-stirred reactor tank" assumption)

⑤

We can write:

① $\dot{V}_{p,out} = \frac{V_{p,sys}}{30,000 \text{ s}} \times 600 \text{ l}^3/\text{min} = .02 V_{p,sys} (\text{l}^3/\text{min})$ where, $V_{p,out}$ = volume flow rate of d-phen out of the cabin (inside of the exiting stream)

$V_{p,sys}$ = volume of d-phen inside the cabin chamber @ time t

And now, we can finally write our Diff. E. Q.

$\frac{dn_{p-sys}}{dt} = -\dot{n}_{p,out}$

← moles d-phen in the cabin and that quantity's rate of change over time

← the moles of d-phen that leave with the exiting stream

Next, we do a trick: → Assume that I.G.L. is valid... then, $PV = nRT$ & $V = \frac{nRT}{P}$

$$\left[\frac{dn_{p-sys}}{dt} = -\dot{n}_{p,out} \right] \cdot \frac{RT}{P} \Rightarrow \frac{d \frac{nRT}{P}}{dt} = -\frac{\dot{n}RT}{P}$$

which is the same as $\frac{dV_{p-sys}}{dt} = -\dot{V}_{p-out}$ and we know this value from Eq. marked

so, $\frac{dV_{p-sys}}{dt} = -.02 V_{p-sys} \Rightarrow V_{p-sys} = V_{p-sys}^0 \cdot e^{-.02t}$ I.C.

① above...

$$V_{p-sys} = 300 \text{ l}^3 \text{ d-phen} \cdot e^{-.02t} = 3 \text{ l}^3$$

now, solve for t ($=t_p$) ($t_p = 230 \text{ min}$)

⑥

(3) Murphy 3.29

$$Acc. = in - out + gen^0 - con$$

in: 62,000 molecules/hr

out: 57,000 molecules/hr

con: $2700(1 - e^{-.3t})$ (molecules/hr)

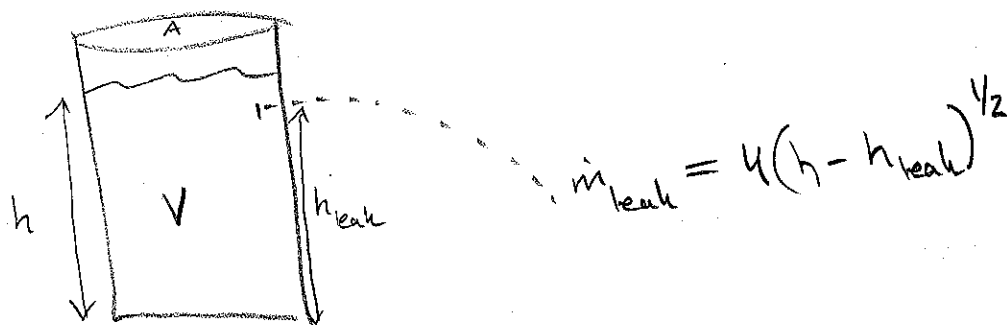
$$\begin{aligned} \therefore Acc.(8hrs) &= 8(62,000 - 57,000) - \int_0^8 2700(1 - e^{-.3t}) dt \\ &= 40,000 - \left[2700(8) + \frac{2700}{.3}(e^{-.3(8)} - e^{(0)}) \right] \\ &= 40,000 - 21600 + 816 - 9000 \\ &= \underline{26,580 \text{ molecules @ } t = 8 \text{ hrs}} \end{aligned}$$

The number is slightly less than the minimum specified in the problem statement.

Your decision should accordingly be not to invest.

⑦

(4) Murphy 3.41



We can solve this problem by balancing compound X

@ the height of the leak, $V_{leak} = \frac{\pi(1m^2)}{4} \times 1m = 0.785 m^3$

from the given density, this is 785 kg sol & 78.5 kg X

$$m_{X@h_{leak}} - m_X^0 = \int_0^{t_{leak}} \dot{m}_{Xin} dt$$

$\dot{m}_{Xin} \leftarrow \text{kg X/m}^3 \times \text{m}^3/\text{min}$

$$78.5 - 0 = 4(t_{leak} - 0)$$

$$t_{leak} = 19.6 \text{ min}$$

after this time, there is an outflow:

$$\dot{m}_{Xout} = .4\sqrt{h-1}$$

we write:

$$\frac{dm_X}{dt} = 4 - .4\sqrt{h-1} = .1 \text{ kg X/kg} \left(\frac{\pi}{4} m^2 \right) \times 1000 \text{ kg/m}^3 \frac{dh}{dt}$$

⑧

Simplifying gives: $78.5 \frac{dh}{dt} = (4 - .4\sqrt{h-1})$

$$78.5 \int_1^{h_p} \frac{dh}{(4 - .4\sqrt{h-1})} = \int_{19.6}^{40} dt = 20.4 \text{ min}$$

Solving (either analytically or numerically) for h_p we get $h_p = 1.975 \text{ m}$

The total amt. of X pumped into the tank $\underbrace{(19.6 \leq t \leq 40)}_U$:

$$.1 \text{ kg X/kg} \cdot 40 \text{ kg/min} \cdot (40 - 19.6) \text{ min} = 81.6 \text{ kg X}$$

The net amount of X added from 1m to 1.975m

$$15 \quad (1.975 - 1) \text{ m} (\pi/4) (1 \text{ m})^2 (1000 \text{ kg/m}^3) (.1 \text{ kg X/kg}) = 76.5 \text{ kg}$$

Amt. of X on the floor U

$$(81.6 - 76.5) \text{ kg X} = \boxed{5.1 \text{ kg X}}$$

''

